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PROPOSED FOR
Accounting Information System Subject

DIGITAL TRANSFORMATION OF CV. MITRA TANI (AGRI-DISTRIBUTOR) CASE STUDY

Analysis of the Expenditure Cycle and Internal Control Based on Accounting Information System (AIS) Review

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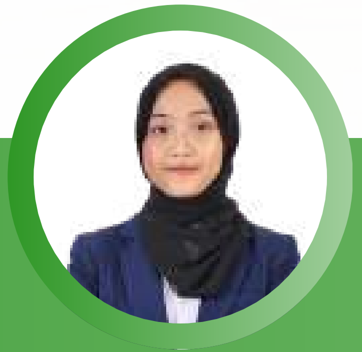

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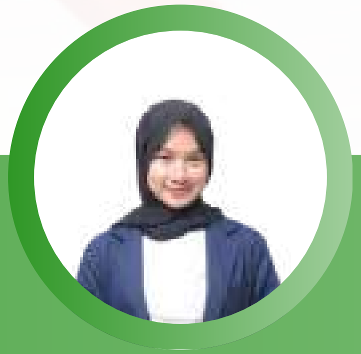

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Contents of The Report

Company Background

CV. Mitra Tani

CV. Mitra Tani is an agricultural distribution company that connects mountain farmers with institutional buyers such as supermarkets and hotels. The company operates a warehouse facility where fresh produce is received, sorted, graded into Quality A, B, and C, repacked according to customer specifications, and stored in cold storage before distribution. As a business-to-business (B2B) distributor, CV. Mitra Tani plays an important role in ensuring product quality consistency, timely delivery, and reliable supply within the fresh produce supply chain.

Currently, the company relies on semi-manual operational systems. Customer orders are received through WhatsApp, while transaction records and inventory data are maintained using Microsoft Excel. Warehouse documentation, including receiving and grading records, is handled separately and is not integrated into a centralized system. As transaction volumes and operational complexity increase, this fragmented system structure highlights the growing need for a more integrated and structured Accounting Information System to support efficiency, transparency, and sustainable business growth.

PROBLEM ANALYSIS

1

Revenue Cycle
Order processing is informal (WhatsApp-based) and not system-integrated. There is no real-time inventory validation, resulting in delayed confirmation, inaccurate stock data, and weak audit trails.
Impact: Reduced sales efficiency and unreliable revenue reporting

2

Expenditure Cycle
Supplier relationships are informal, and harvest records are manually separated from payment data. Payments are not linked to grading results, causing inaccurate cost allocation.
Impact: Distorted COGS and limited cost transparency

3

Conversion Cycle (Critical Risk)
There is no input–output tracking or shrinkage monitoring. Yield cannot be measured, supplier performance cannot be evaluated, and profit margins cannot be accurately calculated.
Impact: Hidden losses and unclear profitability

References: Grande, E. U., Estébanez, R. P., & Colomina, C. M. (2011).

PROCESS MAPPING (DFD Level 0)

2.Data Stores

Data Store Name	Description
Farmers	Master data of suppliers
Customers	Master data of buyers
Products	List of traded commodities
Grades	Quality classification (A, B, C)
Harvest Receipt	Record of goods received
Warehouse Batch	Processing lot control
Sorting & Grading Record	Output quantities by grade and waste
Inventory	Available stock by product and grade
Sales Order	Customer transaction records
Farmer Payment	Payable amount to farmers
Customer Payment	Receivable payments from customer
Accounting Transaction	Journal entries for financial reporting

1.External Entities

External Entity	Role in System
Farmer	Supplier of agricultural
Customer	Buyer
Management	Decision maker

References:(Kumar & Singh, 2022; Helo & Hao, 2023).

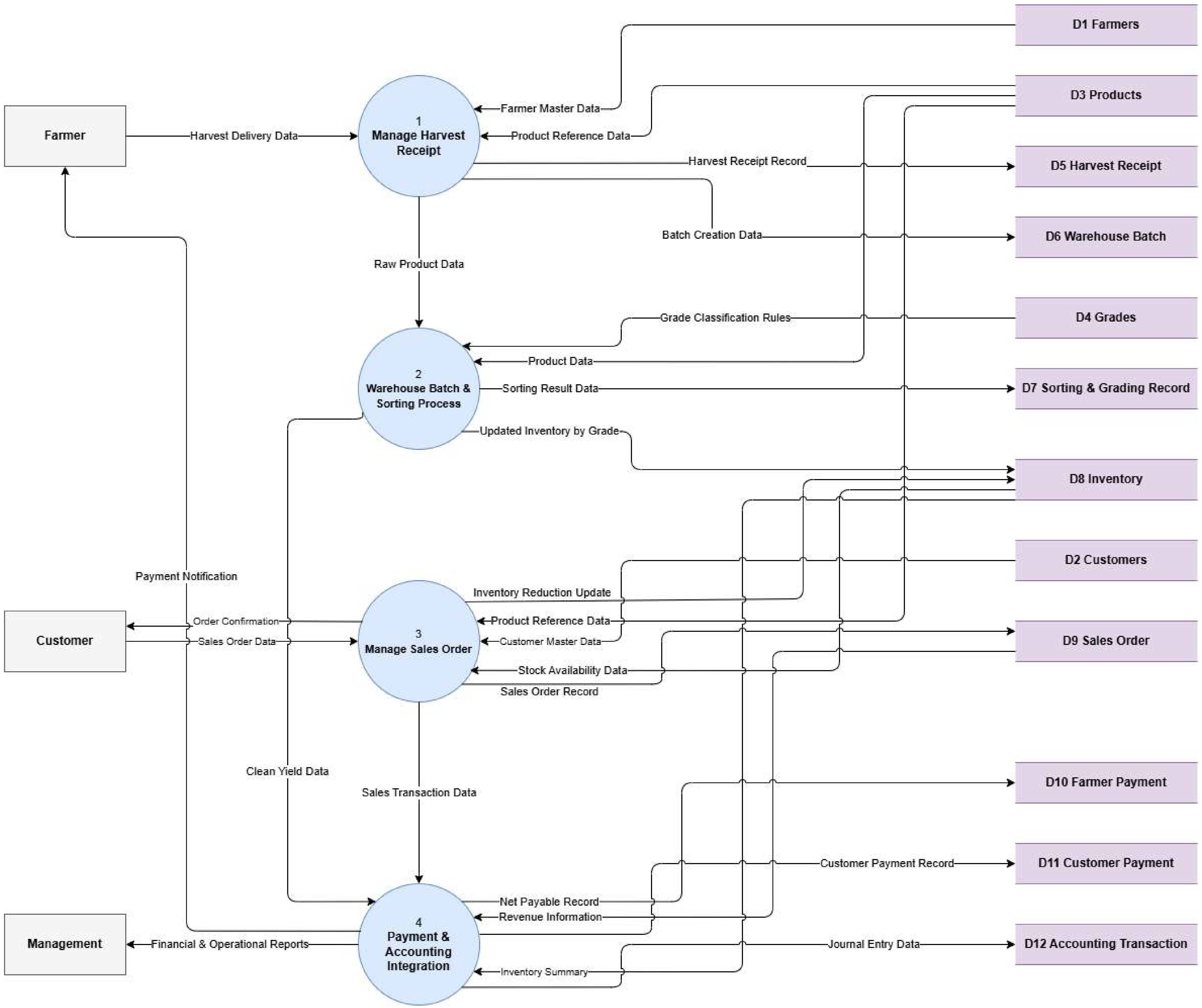
PROCESS MAPPING (DFD Level 0)

3.Major Processes

Process Name	Description	Related Business Cycle
Manage Harvest Receipt	Records product deliveries from farmers and generates official receiving documentation. Initiates traceability for batch processing.	**Expenditure Cycle**
Warehouse Batch & Sorting Process	Controls processing batches, performs sorting & grading (A/B/C), records waste, and updates inventory.	**Conversion Cycle**
Manage Sales Order	Processes customer purchase orders and allocates available inventory by product and grade.	**Revenue Cycle**
Payment & Accounting Integration	Calculates farmer payments based on clean yield, records customer payments, and generates journal entries for financial reporting.	**Financial Integration**

References:(Kumar & Singh, 2022; Helo & Hao, 2023).

Diagram DFD Level 0



References:(Kumar & Singh, 2022; Helo & Hao, 2023).

DATA MAPPING & ENTITY IDENTIFICATION

Data Mapping & Entity Identification defines the structured data architecture required in the proposed integrated Accounting Information System (AIS) of CV. Mitra Tani.

Currently, the company relies on fragmented Excel spreadsheets and WhatsApp communication, resulting in data inconsistency, lack of inventory visibility, and inaccurate profit measurement. Therefore, a structured entity-based database is necessary to integrate operational and accounting information.

This data design supports:

- Accurate inventory tracking by grade (A, B, C)
- Transparent farmer performance evaluation
- Automated farmer payment calculation
- Real-time stock availability
- Integrated revenue, expenditure, and conversion cycles
- Reliable profit margin analysis

Entity Classification by Business Cycle

To ensure systematic integration, data entities are classified based on the three core business cycles.

1. **Expenditure Cycle (Farmer→Warehouse)**

ENTITY	FUNCTION
Farmers	Supplier master data
Harvest Receipt	Record of goods received
Warehouse Batch	Processing lot control
Farmer Payment	Payment to suppliers

The Expenditure Cycle entities manage the procurement process from farmers.

Farmers serve as supplier master data. Harvest Receipt records each delivery received. Warehouse Batch controls processing traceability, while Farmer Payment ensures that supplier compensation is calculated and documented accurately.

This structure ensures that every harvest can be traced from receipt to payment.

DATA MAPPING & ENTITY IDENTIFICATION

2. **Conversion Cycle (Warehouse Processing)**

ENTITY	FUNCTION
Sorting & Grading Record	Record output by grade
Grades	Quality classification
Inventory	Ready-to-sell stock

The Conversion Cycle represents the core operational process. Sorting & Grading Record captures output quantities by quality grade and waste. Grades define quality standards, and Inventory stores ready-to-sell stock by product and grade.

This structure allows management to measure shrinkage, processing efficiency, and actual yield after sorting.

3. **Revenue Cycle (Warehouse → Customer)**

ENTITY	FUNCTION
Customer	Buyer master data
Sales Order	Order header
Sales Order Detail	Order line items
Customer Payment	Payment received

The Revenue Cycle entities manage the sales process. Customer stores buyer master data. Sales Order and Sales Order Detail record transaction details, while Customer Payment tracks incoming payments.

By linking Sales Order Detail with Inventory by grade, the system prevents overselling and ensures accurate revenue recording.

DATA MAPPING & ENTITY IDENTIFICATION

4. **Accounting Integration**

ENTITY	FUNCTION
Accounting Transaction	Journal entry for financial reporting

Accounting Transaction integrates all operational activities into the financial system. Every purchase, sale, payment, and receipt automatically generates journal entries.
This ensures real-time financial reporting and eliminates manual re-entry errors.

MASTER DATA ENTITIES

Master data contains relatively stable information used repeatedly across transactions.

1. **Farmers**

ATTRIBUTE	FUNCTION	
Farmer_ID	Primary Key	Uniques farmer code
Farmer_Name		Full name
Phone_Number		contact
Address		Location
Bank_Account_Number		Payment account
Contract_Status		Contracted/ Non-Contracted
Performance_Rating		Based on grading history

DATA MAPPING & ENTITY IDENTIFICATION

2. **Customers**

ATTRIBUTE	FUNCTION	DESCRIPTION
Customer_ID	PK	Unique customer ID
Customer_Name		Business name
Business_Type		Hotel / Supermarket
Contact_Person		Person in charge
Phone_Number		Contact
Address		Delivery location
Credit_Terms		Payment agreement

The Customers table stores key information about clients, including business identity, business type, contact details, and payment terms. This table is essential for ensuring controlled sales, accurate payment recording, and that customer data is ready for business analysis and financial integration.

3. **Products**

ETTRIBUTE	FUNCTION	DESCRIPTION
Product_ID	Primary Key	Unique product code
Product_Name		Produce name
Unit_of_Measure		Kg / Box
Standard_Purchase_Price		Estimated cost
Standard_Selling_Price		Base selling price
Storage_Type		Cold / Dry

The Products table defines each commodity traded by the company. It includes unit measurement, standard purchase price, standard selling price, and storage requirements.
This entity supports costing, pricing decisions, and warehouse management.

DATA MAPPING & ENTITY IDENTIFICATION

4. **Grades**

ATTRIBUTE	FUNCTION	DESCRIPTION
Grade_ID	PK	A / B / C
Grade_Name		Quality label
Description		Quality criteria
Price_Adjustment		% difference from base price

The Grades table defines quality classifications (A, B, C) including pricing adjustments. By separating grades from products, the system enables dynamic pricing based on quality level and supports profit margin analysis by grade.

TRANSACTIONAL ENTITIES

Transactional entities record operational activities.

1. **Harvest Receipt**

ETTRIBUTE	FUNCTION	DESCRIPTION
Receipt_ID	Primary Key	Unique receipt
Farmer_ID	Foreign Key	Linked to Farmers
Product_ID	Foreign Key	Product delivered
Date_Received		Receiving date
Gross_Weight		Total weight before sorting
Initial_Quality_Notes		Inspection notes
Receiving_Staff_ID		Responsible staff

DATA MAPPING & ENTITY IDENTIFICATION

2. **Warehouse Batch**

ATTRIBUTE	FUNCTION	DESCRIPTION
Batch_ID	Primary Key	Processing lot number
Receipt_ID	Foreign Key	Linked to Harvest Receipt
Processing_Date		Sorting date
Batch_Status		Pending / Completed

Warehouse Batch controls processing activities using lot numbers. One Harvest Receipt may be processed into multiple batches to maintain traceability and operational control. Batch tracking is essential for quality management and shrinkage analysis.

3. **Sorting & Grading Record**

ATTRIBUTE	FUNCTION	DESCRIPTION
Sorting_ID	Primary Key	Unique sorting record
Batch_ID	Foreign Key	Linked to Warehouse Batch
Grade_A_Weight		Output A
Grade_B_Weight		Output B
Grade_C_Weight		Output C
Waste_Weight		Shrinkage
Staff_ID		Sorting staff

DATA MAPPING & ENTITY IDENTIFICATION

4. **Inventory**

ATTRIBUTE	FUNCTION	DESCRIPTION
Inventory_ID	Primary Key	Unique ID
Product_ID	Foreign Key	Product reference
Grade_ID	Foreign Key	Grade reference
Quantity_Available		Current stock
Last_Updated		Timestamp

Inventory stores available stock by product and grade. It is automatically updated by sorting results and reduced by sales orders. This enables real-time stock visibility and eliminates manual warehouse confirmation.

5. **Sales Order**

ATTRIBUTE	FUNCTION	DESCRIPTION
Sales_Order_ID	Primary Key	Order number
Customer_ID	Foreign Key	Buyer
Order_Date		Order date
Order_Status		Pending / Delivered

Sales Order records customer purchase transactions at a summary level, including order date and status. It initiates revenue recognition and inventory allocation.

DATA MAPPING & ENTITY IDENTIFICATION

6. **Sales Order Detail**

ATTRIBUTE	FUNCTION	DESCRIPTION
Order_Detail_ID	Primary Key	Line item ID
Sales_Order_ID	Foreign Key	Linked to order header
Product_ID	Foreign Key	Product ordered
Grade_ID	Foreign Key	Grade ordered
Quantity		Amount
Unit_Price		Selling price
Subtotal		Quantity × Price

Sales Order Detail records the specific product, grade, quantity, and price ordered. This level of detail ensures accurate billing, inventory reduction, and margin calculation per quality grade.

7. **Farmer Payment**

ATTRIBUTE	FUNCTION
Farmer_Payment_ID	Primary Key
Farmer_ID	Foreign Key
Receipt_ID	Foreign Key
Net_Payable_Amount	
Payment_Date	
Payment_Status	

Farmer Payment records the net payable amount to each farmer based on clean yield after sorting. By linking payment to actual grading results, the system increases transparency and fairness.

DATA MAPPING & ENTITY IDENTIFICATION

8. **Customer Payment**

ATTRIBUTE	FUNCTION
Customer_Payment_ID	Primary Key
Sales_Order_ID	Foreign Key
Amount_Paid	
Payment_Date	
Outstanding_Balance	

The Customer Payment table records all payments received from customers for sales transactions. It tracks payment amounts, payment dates, and outstanding balances. This table ensures proper receivable management, improves cash flow visibility, and maintains transparency by linking payments to specific sales orders.

9. **Accounting Transaction**

ATTRIBUTE	FUNCTION
Transaction_ID	Primary Key
Transaction_Date	
Reference_Type	
Reference_ID	
Debit_Amount	
Credit_Amount	
Account_Code	

The Accounting Transaction table records the financial impact of all operational activities, including purchases, sales, and payments. Each entry is linked to its originating transaction, ensuring accurate journal entries, real-time financial reporting, and a clear audit trail for transparency and control.

Overview of the Receiving Function

According to Hall (2016), effective transaction processing systems must incorporate input validation, authorization controls, and documentation procedures at the point of data entry. Therefore, the redesigned Receiving Process is structured to:

1. Formally recognize delivery from farmers
2. Capture Gross Weight as the quantitative basis of procurement
3. Automatically generate a unique Batch_ID for traceability
4. Integrate operational data with accounting records

The Receiving Process thus functions not only as an operational checkpoint but also as a control mechanism ensuring the integrity of downstream sorting, inventory updates, and payment calculation.

Digital Standard Operating Procedure (SOP) - Receiving Process

Step 1: Farmer Identification and Delivery Verification

The system retrieves:

- Farmer Master Data (D1)
- Product Reference Data (D3)

Step 2: Digital Weighing and Gross Weight Recording

- Gross_Weight (Kg)
- Date_Received
- Time_Received

Step 3: Preliminary Inspection

- Visible spoilage
- Packaging condition
- Moisture condition
- General product appearance

Digital Standard Operating Procedure (SOP) - Receiving Process

Step 4: Harvest Receipt Creation

- Receipt_ID (Primary Key)

Step 5: Automatic Batch_ID Generation (Traceability Mechanism)

- Batch_ID
- Example format:
- BATCH-YYYYMMDD-XXX

- The Batch_ID serves as the integration bridge between:
- Expenditure Cycle (procurement)
- Conversion Cycle (sorting and grading)

Data Recording Requirements

Attribute	Description	Control Principle
Receipt_ID	Auto-generated primary key	System control
Farmer_ID	Foreign key to Farmers	Master data validation
Product_ID	Foreign key to Products	Reference validation
Gross_Weight	Total delivered weight	Automated input control
Date_Received	Timestamp	System-generated
Initial_Quality_Notes	Inspection notes	Documentation control
Receiving_Staff_ID	User identification	Accountability
Batch_ID	Processing lot number	Traceability

System Integration

Integration with Conversion Cycle. The Receiving Process transmits the following data to the Sorting Process:

- Batch_ID
- Gross_Weight
- Product_ID
- Farmer_ID

FATIMAH AZZAHRA

Integration with Expenditure Cycle
After sorting is completed, Clean Yield is calculated as:

Clean Yield = Grade A + Grade B + Grade C

The Clean Yield is linked back to:

- Receipt_ID
- Farmer_ID

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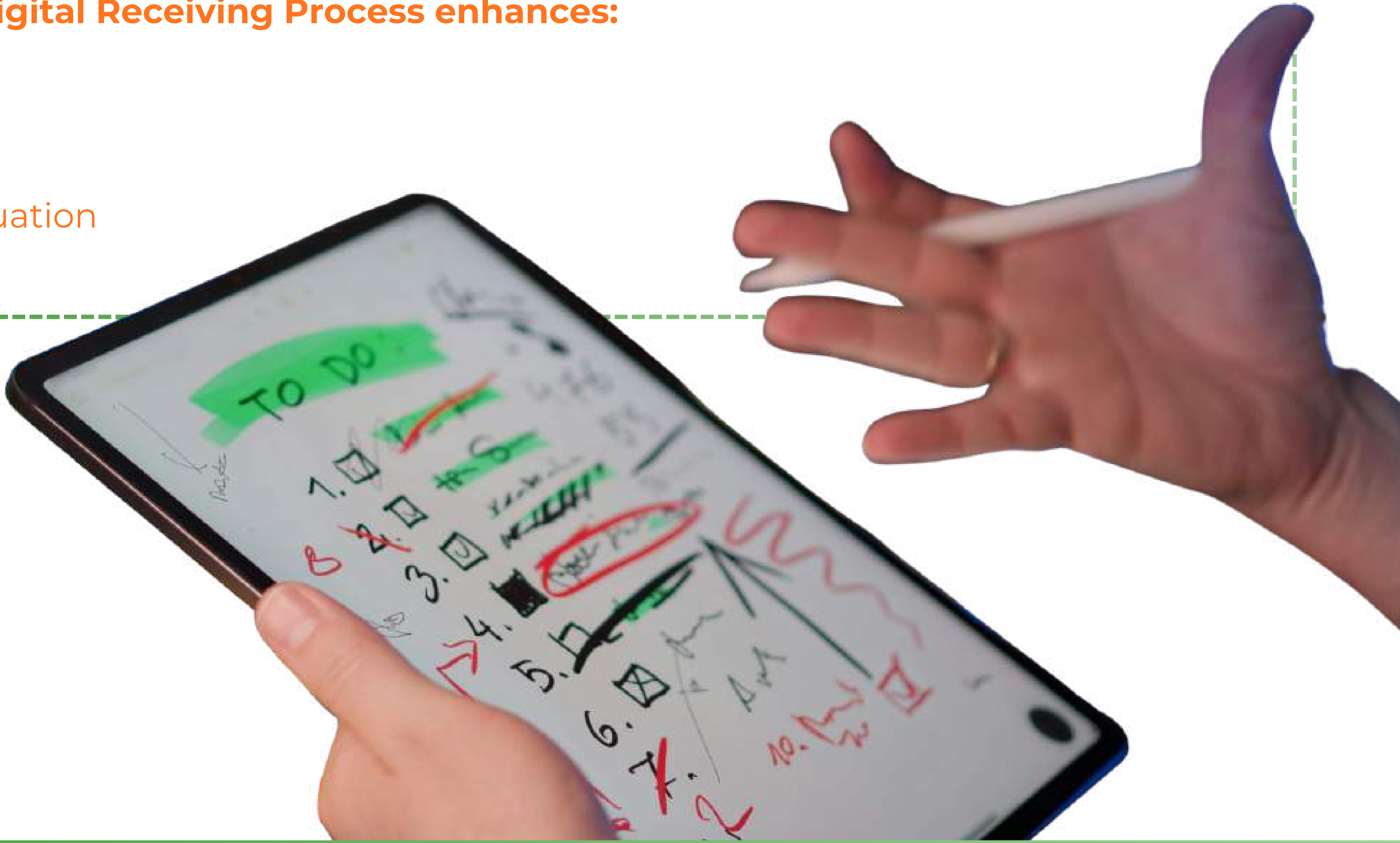
Strategic Implications

Implementing a structured digital Receiving Process enhances:

- Transaction completeness
- Data reliability
- Inventory traceability
- Supplier performance evaluation
- Profit margin accuracy



Risk and Internal Control Analysis

Risk	Impact	Recommended Control
Overstated Gross Weight	Overpayment to farmers	Digital scale integration
Unauthorized supplier transaction	Fraud risk	Mandatory Farmer_ID validation
Duplicate receipt entry	Financial distortion	Auto-generated Receipt_ID
Data manipulation	Integrity loss	Role-based access control
Missing Batch_ID	System breakdown	Mandatory system-generated batch

Core System Design Sorting, Grading & Repacking

Overview of the Conversion Process

- The conversion cycle transforms heterogeneous raw agricultural inputs into homogeneous outputs based on strict quality parameters
- Core Function: It acts as a critical financial gateway to accurately measure the transition from Gross Weight to Clean Yield.

Digital Standard Operating Procedure (SOP)

- 1. Batch Initiation: Staff scans the Batch_ID barcode attached to the raw material pallet to retrieve the Gross_Weight
- 2. Physical Sorting: Produce is physically separated into distinct containers based on company standards:
  - Grade A: Premium quality, no physical defects.
  - Grade B: Standard quality, minor variations.
  - Grade C: Sub-standard/Processing quality.
  - Waste: Shrinkage or spoilage.

Algorithmic Validation & Data Requirements

Data Recording RequirementsTo maintain database integrity, the digital input screen captures:

- Primary Key: Sorting_ID
- Foreign Keys: Batch_ID (from Receiving), Staff_ID (for audit trail)
- Yield Metrics: Grade_A_Weight, Grade_B_Weight, Grade_C_Weight, Waste_Weight

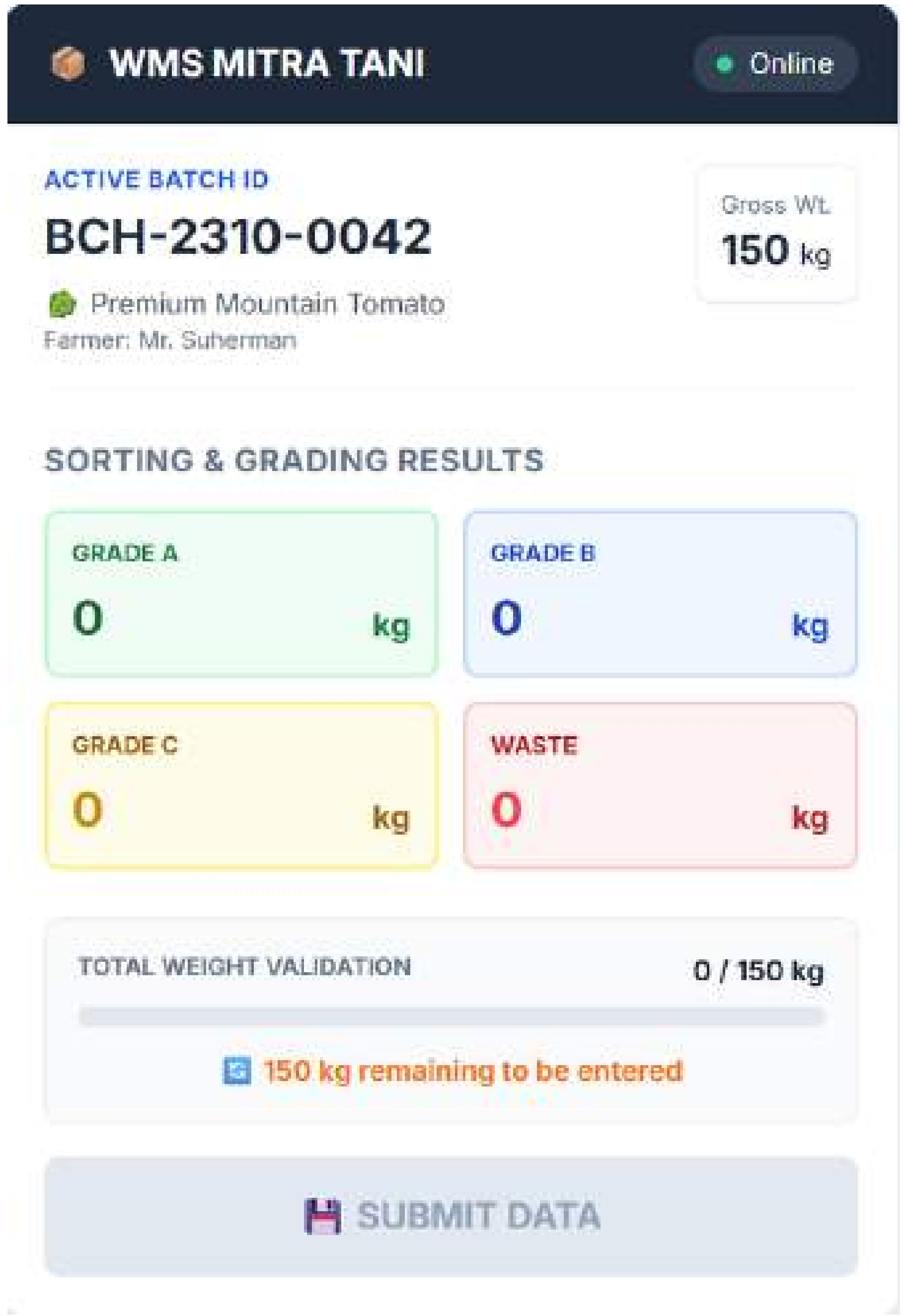
Internal Control Point: Algorithmic Validation

- The system performs automatic validation to prevent lost items or recording errors.
- The "Submit" button remains **locked** (disabled) until the mathematical equation is fulfilled:
- **Gross Weight = Grade A + Grade B + Grade C + Waste**

UI/UX Design & Internal Control Automation

- The input form is built using a Mobile-First approach to minimize Human Error.
- Features Psychological Color-Coding to prevent "fat-finger errors" and a Dynamic Progress Bar as a visual indicator

Attribute	Data Requirement
Batch ID	BCH-2310-0042
Gross Weight	150 Kg
Grade A Input	[______] Kg
Grade B Input	[______] Kg
Grade C Input	[______] Kg
Waste Input	[______] Kg
Status	Validation Check



Repacking & Inventory Blueprint

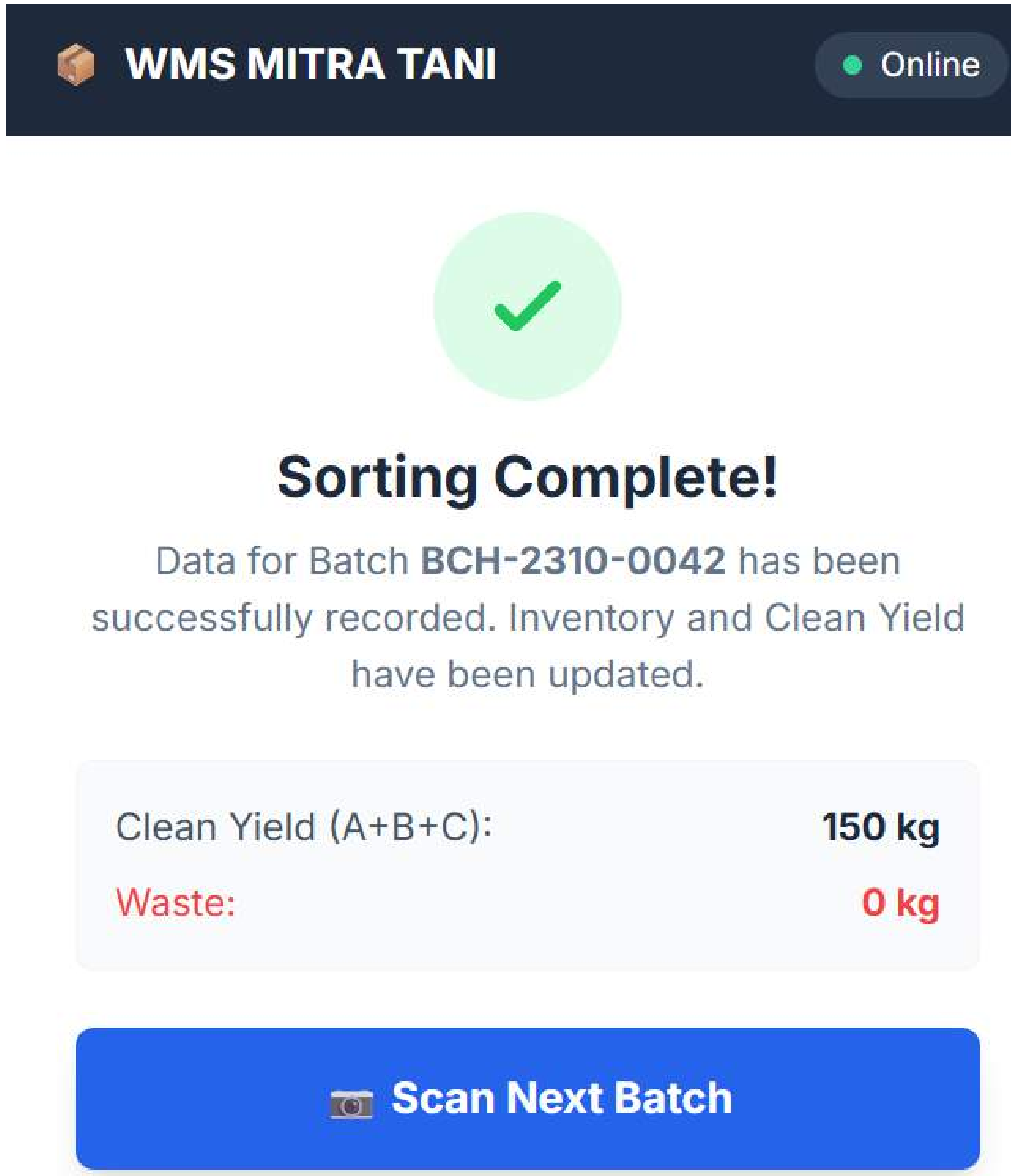
Post-Validation System TriggersOnce the data is validated and submitted:

- Inventory Updated: The system instantly updates ready-to-sell inventory for the Revenue Cycle.
- Clean Yield Sent: Grade A + B + C data is sent to the finance department to calculate accurate farmer payments.
- Repacking & Labeling: The system automatically prints a new Finished Goods barcode for each repacked box.
- End-to-End Traceability: The new barcode is relationally linked to the original Batch_ID.
- Offline-First Architecture: Data is stored in the local cache to accommodate Cold Storage facilities with poor internet signals

System Action	Data Output / Status
Batch ID	BCH-2310-0042
Clean Yield (A+B+C)	150 Kg (Ready to Pack)
Waste / Shrinkage	0 Kg (Disposed)
Repacking Size	5 Kg & 10 Kg Boxes
Inventory Status	+150 Kg Added to System
Traceability	New Barcodes Generated

Post-Validation Triggers: Automatically updates inventory and calculates Clean Yield for finance.

Automated Success Prototype



SYSTEM INTEGRATION (REVENUE & EXPENDITURE CYCLE)

Overview of System Integration

The proposed Accounting Information System (AIS) integrates the Conversion, Revenue, and Expenditure Cycles in one centralized database. The warehouse sorting and grading process becomes the main integration point because it determines ready-to-sell inventory and the clean yield used for farmer payments.

SYSTEM INTEGRATION (REVENUE & EXPENDITURE CYCLE)

Integration with Revenue Cycle (Ready-to-Sell Stock)

In the proposed AIS, warehouse conversion data directly determines ready to sell inventory in the Revenue Cycle.

- **Integration Mechanism:**
- Harvest receipt recorded
- Sorting & grading produce Grade A, B, C
- Grading results automatically update inventory
- Sales order → system checks real-time stock
- Confirmed order → stock reduced & revenue recorded automatically
    -

Through this process, warehouse operational data is directly connected to the sales process, ensuring that inventory information in the Revenue Cycle always reflects actual graded output.

- **Key Impact:**
- Real-time stock visibility per grade
- Prevents overselling
- Accurate quality-based sales allocation

SYSTEM INTEGRATION (REVENUE & EXPENDITURE CYCLE)

Expenditure Cycle Integration

In the integrated AIS, warehouse operational data is also connected to the Expenditure Cycle to ensure objective and accurate farmer payment calculation. The payment is not based on total gross delivery, but on the actual usable output after the sorting and grading process

.

Clean Yield Rule:

- Gross Weight = Grade A + B + C + Waste
- Clean Yield = Grade A + B + C
- Farmer payment is based only on Clean Yield

Integration Mechanism:

- Harvest receipt records total delivery (gross)
- Sorting identifies quantity per grade & waste
- System calculates clean yield automatically
- Clean yield becomes basis for farmer payment
- Payment transaction recorded in Expenditure Cycle

Key Impact:

- Objective and transparent farmer payment calculation
- Reduced disputes regarding quantity and quality
- Accurate expenditure recording in the system
- Better financial control and accountability

CV. MITRA TANI

INTERNAL RISK MANAGEMENT

THE CONTROL PARADOX

Digital Benefits	New Risks Introduced
85% reduction in human error	Concentrated system access
Automated validation	Override opportunities
Real-time integration	Single point of failure
Complete audit trail	Data manipulation risks

Digital transformation shifts risk from human error to system vulnerability

KEY INDICATOR	RISK	OPPORTUNITIES
CLIMATE	Increased operational costs due to carbon regulations	Shift to renewable energy lowers long-term expenses
REPUTATION	Negative brand image from poor environmental practices	ESG transparency builds customer trust and loyalty
COMPLIANCEMATE	Fines from failing to meet environmental standards	Early compliance gives first-mover advantage

BORCELLE COMPANY

RISK HEAT MAP CONVERSION CYCLE FOCUS

CONCLUSION

CV. Mitra Tani main issue lies in the lack of an integrated Accounting Information System across its revenue, expenditure, and conversion cycles. Semi-manual processes and fragmented data reduce accuracy, transparency, and operational efficiency. The absence of real-time inventory validation, yield tracking, and cost integration limits the company’s ability to control costs, evaluate supplier performance, and determine accurate profit margins.

To support sustainable growth, the company requires a more structured and integrated AIS that ensures reliable data, stronger internal controls, and better decision-making capability.

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Thank You